Abu Dhabi City Campus Daily/Weekly Planner مجمع مدينة أبوظبي -الخطة اليومية الإسبوعية

Week #11 : Ecology GRASPS Performance Task & Symbiosis

Subject/الموضوع: Pre-AP Biology Unit Title: Ecological Systems Grade9: الصف

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1	I can plan a fair comparison of two conservation actions in a drought-affected UAE reserve. I can collect and record original virtual-lab results for vegetation, herbivores and predators.	1.1 Flexible display of information (size, color, font, layout).	Grade 9 Ecology GRASPS Task — Day 1: investigation question, prediction, 4–6 step plan, custom ENS Ecology Conservation Virtual Lab, two trials (vegetation restoration vs predator protection), Year 0/2/4/6/8 data and screenshots. Handwritten evidence sheet; device; Teams task.
Lesson 2 الحصة الثانية	I can calculate population changes and represent my own simulation data in a labeled graph. I can use ecological concepts and numerical evidence to justify a UAE conservation recommendation.	5.3 Build fluency with graduated scaffolds and models.	Grade 9 Ecology GRASPS Task — Day 2: handwritten results table, herbivore-change calculations, line graph, four-organism UAE food web, limiting factors/carrying capacity, CER recommendation, model limitation and trade-off; 6–8-slide evidence briefing.
Lesson 3 الحصة الثالثة	I can defend a conservation decision using my own simulation data and ecological reasoning. I can evaluate the strengths and limitations of ecological models and reflect on my investigation.	6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides.	Grade 9 Ecology GRASPS Task — Day 3: individual 3-minute conservation briefing, live scientific question, original handwritten evidence, digital slides and final reflection. MYP Sciences task-specific criteria A–D (each /8); presentation panels if needed.
Lesson 4 الحصة الرابعة	I can classify mutualism, parasitism and commensalism from evidence of how each species is affected. I can explain the benefits, costs or lack of measurable effects for both partners.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	Lesson 20: Symbiosis — slides 1–8; warm-up (+/–/0), evidence-pair investigation, mutualism, parasitism and commensalism, answer key and 1-minute peer feedback; notebook and PowerPoint.
Lesson 5 الحصة الخامسة	I can explain how symbiotic relationships influence survival and access to resources. I can predict community consequences when a symbiotic partner declines, using evidence and a cause–effect model.	8.2 Vary challenge level with differentiated tools, resources, and difficulty.	Lesson 20: Symbiosis — slides 9–13; UAE coral reef cleaner-fish case, cause–effect model, peer-feedback rubric, AP-style questions, exit ticket and extension; PowerPoint and notebook.

Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	• GRASPS lessons: bring a device for the virtual ecology lab and briefing slides, plus paper for compulsory handwritten planning, data, graph and CER. • Keep both trial settings identical except the conservation action; retain screenshots and original readings. • Plan presentation panels for larger classes (approximately 4 minutes per student including question). • Symbiosis lessons: notebook, projected evidence tables, peer-feedback checklists and UAE coral reef case study.

الأسبوع# Week #11

Technology Integration دمج التكنولوجيا	• GRASPS: custom ENS Ecology Conservation Virtual Lab linked in the uploaded task; screenshots of Trials A/B; digital 6–8-slide briefing. • Lesson 20: PowerPoint models, evidence tables, answer keys and peer-feedback slides. • Microsoft Teams for access to learning resources and task instructions.
Cross-curricular links الروابط بين المواد الدراسية	• Mathematics: calculate Year 0 to Year 8 changes; draw and interpret a two-series line graph; compare percentages and patterns. • Geography/Environmental Science: drought, habitat restoration, predator protection and UAE reserve management. • English: CER writing, oral defense and evidence-based explanations.
External Assessment Integration دمج التقييمات الخارجية	Pre-AP Biology Unit 1 — Ecological Systems: ECO 1/ECO 2/ECO 4 synthesis in GRASPS and ECO 4.2 symbiosis. Apply scientific models, interpret evidence, construct explanations and justify conservation decisions. GRASPS uses the attached MYP Sciences task-specific A–D rubric (/32); Lesson 20 includes AP-style transfer questions.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	• Thinking: fair-test design, systems thinking, data analysis, causal reasoning and evidence evaluation. • Research: accurate recording of simulation outputs and identification of model limitations. • Communication: CER writing, scientific presentation, responding to questions and peer feedback. • Self-management: planning, completing original work and reflecting on feedback.
Standards المعايير	• Lessons 1–3: Ecology GRASPS — integrated ecology concepts: matter/energy, populations, limiting factors, carrying capacity, food webs, biodiversity and conservation decisions; MYP Sciences Criteria A–D applied as specified in the task. • Lessons 4–5: Pre-AP ECO 4.2(a–b), EK 4.2.1a–c — mutualism, parasitism and commensalism.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	Lesson 1: How can we fairly compare vegetation restoration and predator protection during drought? Lesson 2: Which conservation action is better supported by the simulation data, and why? Lesson 3: How reliable is a model when making real conservation decisions? Lesson 4: How can evidence show whether each partner benefits, is harmed or is unaffected? Lesson 5: How can losing one symbiotic partner affect a wider ecological community?
Success Criteria معايير النجاح **(Measurable)**	Lesson 1 — GRASPS Plan & Test MUST (DOK 1): Identify the two actions and record results for both trials. SHOULD (DOK 2): Write a repeatable fair-comparison plan and prediction. COULD (DOK 3): Justify controlled settings and predict ecological effects. Lesson 2 — GRASPS Analyze & Decide MUST (DOK 1): Complete results table and calculate herbivore change. SHOULD (DOK 2–3): Draw an accurate graph and explain food-web/carrying-capacity effects. COULD (DOK 4): Defend a CER recommendation using two data values, a limitation and a trade-off. Lesson 3 — GRASPS Defend MUST (DOK 1): Present own data and conclusion. SHOULD (DOK 2–3): Explain evidence and respond to a scientific question. COULD (DOK 4): Evaluate model limitations and conservation impacts.

	Lesson 4 — Symbiosis Types MUST (DOK 1): Identify mutualism (+/+), parasitism (+/−) and commensalism (+/0). SHOULD (DOK 2): Classify relationships using evidence for both partners. COULD (DOK 3): Explain uncertainty or a change in interaction outcomes. Lesson 5 — Symbiosis in Communities MUST (DOK 1): Identify benefits in cleaner-fish mutualism. SHOULD (DOK 2): Explain effects of partner decline using a causal model. COULD (DOK 3): Justify community predictions and assess alternative outcomes.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Respect الاحترام
	المجال Area: Conservation / Sustainability / Community
	• • UAE desert reserve: compare restoring vegetation with protecting predators under drought using the custom virtual lab (fictional model data). • UAE conservation committee: weigh ecological evidence, practical trade-offs and model limitations. • UAE coral reefs: examine cleaner-fish mutualism and consequences for reef-community health.